Implementation Models for Data Science

We designed our lessons in Bootstrap:Data Science to be incredibly flexible, with options for teachers in multiple subjects from grade 6-12.

Choose which of the following implementation models is right for you:

Dedicated Data Science Class *1 semester to Full Year*

The Bootstrap:Data Science lessons can be broken up into five distinct units, each with a clear theme and a set of creative projects.

Note: Many teachers pace themselves in order to have their students analyze *additional datasets* after completing all five units. Analyzing multiple datasets allows students to refine their skills and broaden and/or deepen domain knowledge. Most teacher start with one of the provided datasets (or a cleaned one they found themselves), and only have students collect their own data after working through their first analysis.

Unit 1: Who are Data Scientists, and what do they do?

Students see the wide range of people involved with mathematics, computing, and data science, and confront the challenge of answering messy questions with data. They explore a sample dataset, consider the relationship between probability and statistics, and learn the basic programming and statistics necessary to display that dataset using a variety of charts, plots and graphs.

- Computing Needs All Voices
- Introduction to Data Science
- Simple Data Types
- Contracts
- Bar and Pie Charts
- The Data Cycle
- Probability, Inference, and Sample Size

Project Options:

- Making an Infographic
- Food Habits
- Time Use

Unit 2: Gathering, Analyzing, and Visualizing Data

Students choose a real dataset, or create their own! They explore this dataset, diving deep into the meaning, use, and interpretation of one-dimensional analyses like histograms, box-plots, measures of center and spread. What do these tools tell us about our data, and when is it appropriate to use one tool over another?

- Choosing Your Dataset
- Histograms
- Visualizing the "Shape" of Data
- Measures of Center
- Box Plots
- Standard Deviation
- Collecting Data

Project Options:

- Design a Survey
- Exploring Your Dataset

Unit 3: Transforming and Playing with Data

Students learn to program *functions*, super-charging their programming arsenals with core algebraic concepts that allow them to create custom visualations that express many dimensions of data. They also learn how to *sort, filter, and transform* their dataset to search for patterns, zoom in on samples, and identify trends within sub-groups of their samples.

- Row and Column Lookups
- Functions Make Life Easier!
- Functions: Contracts, Examples & Definitions
- Custom Scatter Plots
- Table Methods
- Solving Word Problems with the Design Recipe
- Method Chaining
- Defining Table Functions

Project Options:

- Design Recipe Telephone
- Beautiful Evidence (Coming Soon!)

Unit 4: Modeling the World Around Us with Data

In this unit, students apply their newfound programming power to identify *relationships within their data*, and developing linear (and even non-linear!) models to describe those relationships. They generate numerous data displays, and combine them into a library of visualizations and inferences about their data.

- Grouped Samples
- Scatter Plots
- Correlations
- Linear Regression

Project Options:

- Olympic Records

Unit 5: Using Data Responsibly

As full-fledged data scientists, students consider the *social impact* of their work. How do we know that the tools we use to analyze data haven't been tampered with? How do we know to trust the validity of our methods and data-collection? What are the ethical implications of this work? Using the visualization library from the previous unit, students conduct an original research project into their data. In doing so, they confront both analytical and ethical factors in their final report, and discuss the ethical standards that define a responsible data scientist.

- Checking Your Work
- Threats to Validity
- Ethics, Privacy, and Bias

Project Options:

- Making an Infographic
- Project: Write a Research Paper
- Make the slimiest research paper possible!

Other Considerations

What Domain Knowledge do you care about? Do you want your students to focus on climate systems? Economics? Social Studies or History themes? Do you want them to design a survey for their school or neighborhood? What topics are important to your students? What topics are exciting to them? Your answers to these questions will determine the dataset(s) you'll use or collect, which has significant impacts on

engagement, relevance, and inclusion.

Integrate Data Science an Existing Class *4 weeks, up to 1 semester*

A module with programming aimed specifically at transforming tables and data visualation, designed for:

- Statistics teachers
- Modeling-Based Science teachers
- Computer Science teachers looking to teach more programming
- Data Science teachers

This format includes multiple project-based options, including Making an Infographic, Food Habits, Stress or Chill?, and Time Use.

Build a foundation...

Introduction to Data Science

Simple Data Types

Contracts

Bar and Pie Charts

The Data Cycle

Choosing Your Dataset

Functions: Contracts, Examples & Definitions

Row and Column Lookups

Custom Scatter Plots

Table Methods

...then choose what you need

Collecting Data

Computing Needs All Voices

Method Chaining

Solving Word Problems with the Design Recipe

Defining Table Functions

Grouped Samples

Probability, Inference, and Sample Size

Correlations

Linear Regression

Checking Your Work

Threats to Validity

Ethics, Privacy, and Bias

Other Considerations

What Domain Knowledge do you care about? If you're integrating into a Science class, maybe you want students to study data from experiments, or data related to Earth Science or Biological phenomena from the Next Generation Science Standards. If you're integrating into a Social Studies class, maybe you're looking at datasets involving gerrymandering or redlinling. Your answer to this question will determine the dataset(s) you'll use or collect, which has significant impacts on engagement, relevance, and inclusion.

Which Math and Statistics learning goals do you have? The answer to this question will determine which lessons and projects from our library are relevant to you. A middle-school teacher might focus on lessons dealing pie and bar charts, histograms, etc. An Algebra teacher might focus on lessons about defining and composing functions. Meanwhile, a CS teacher might spend time on If-Expressions and conditionals.

Just a taste of Data Science *1 to 4 weeks*

A module with minimial programming, designed for:

- Science teachers who want students to gather data and generate charts for lab reports
- Math teachers who want students to experiment with charts and plots
- History or Social Studies teachers who want students explore census data, voting data, economic data, etc.
- Computer Science teachers who want a small, gentle exposure to Data Science for their students

In addition to whatever project you want your students to do with the data from your class, this format includes optional projects, such as Making an Infographic and Stress or Chill?.

Build a foundation…

Introduction to Data Science

Simple Data Types

Contracts

Bar and Pie Charts

…then choose what you need

Visualizing the "Shape" of Data

Measures of Center

Box Plots

Scatter Plots

Correlations

Linear Regression

Ethics, Privacy, and Bias